claude-windows / 188dd06f-9c99-487c-aa7d-b0bee83c629c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\188dd06f-9c99-487c-aa7d-b0bee83c629c\subagents\workflows\wf_416cc0e5-a35\agent-a14ccd2c8fdd33eb6.jsonl`
- SHA-256: `91d5aadded6cb94c77de73e9dcc0df1e361a23b2a5ccd73f0c9db7a985921f98`
- Source modified: `2026-06-22T09:45:36+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_416cc0e5-a35`
- session_id: `188dd06f-9c99-487c-aa7d-b0bee83c629c`

Transcript

1. user (2026-06-22T09:42:12.648Z)

Review the NIGHTLY REEL-COUNT engine-regression change under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e (a NEW, additive PR). It runs the configured WASB pipeline on a GCP GPU VM nightly, asserts reel-count (==) + quality metrics + cost, emails a report, self-deletes. Files: scripts/nightly_reelcount.py, scripts/nightly_email.py, deploy/gcp/nightly_regression/run_on_vm.sh, deploy/gcp/nightly_regression/launch.sh, deploy/gcp/nightly_regression/register_scheduler.sh, deploy/gcp/nightly_regression/Dockerfile.launcher, eval_baselines/reelcount_manifest.json. Read them under that absolute path. Cite file:line. Report only REAL defects (correctness/safety/security that would bite in production or a regression that escapes), not style. Return raw structured findings.

LENS: BASH SAFETY in run_on_vm.sh, launch.sh, register_scheduler.sh. CRITICAL: does run_on_vm.sh ALWAYS self-delete the VM (cost rule), including when an early step fails? Analyze the trap EXIT (cleanup/self-delete) + trap ERR (fail_email) interaction under 'set -uo pipefail' ? does ERR firing skip the EXIT trap? does 'trap - ERR' at the end break anything? Is BOOT_EPOCH/uptime correct? Are metadata reads robust if a key is absent? In launch.sh: git archive of HEAD, --metadata escaping (could a value with commas/spaces break --metadata=k=v,k2=v2?), zone sweep. In register_scheduler.sh: gcloud run jobs / scheduler correctness, the Scheduler?Run OAuth. Flag anything that could leave a billing GPU VM alive, or silently not run.

2. assistant (2026-06-22T09:42:14.844Z)

I'll conduct a thorough security and correctness review of the nightly reel-count regression engine, focusing on bash safety, cost control, and production robustness. Let me read the key files.

3. user (2026-06-22T09:42:16.253Z)

```
"""Nightly END-TO-END engine regression ? reel-count + quality + cost (the GCP-VM nightly).

Runs the **configured + checkpointed** pipeline (the real WASB GPU detector, as a deployment serves
it) end-to-end on 2-3 short owned golden videos and asserts the **number of reels does not change**
vs a frozen baseline. Unlike ``scripts/nightly_regression.py`` (PR #202 ? a CPU re-score of CACHED
trajectories, the windowing/scorer only), this one runs the *full* pipeline on the *real model*, so it
needs a GPU ? hence it runs on an ephemeral GCP GPU VM (``deploy/gcp/nightly_regression/``), emails the
owner the report, and self-deletes. It is **default-additive**: a tripwire, not the promotion gate.

```

```
What it asserts / reports, per the owner's ask ("quality metrics + regression report + cost"):
  ? REEL COUNT per video ? ``len(play_segments)`` after segmentation; the detected highlight clips.
    Pure-CV / WASB detection is deterministic for a fixed (video, config, weights), so this is asserted
    **EXACT** (with a re-run-once guard for the one known flake: a transient DB-insert-None, see §guard).
  ? QUALITY METRICS per video (when golden labels are provided) ? the R2 tolerance-match block from
    ``backend.eval.rally_seg_eval`` (tol_f1 / f1@0.5 / f1@0.25), compared with a small tolerance.
  ? COST of the run ? VM uptime (or processing wall) × machine $/hr via ``backend.billing`` ? USD + INR.

Extensible pattern (add a video = one manifest row + re-freeze):
  ``eval_baselines/reelcount_manifest.json``  ? the videos (gs:// / local / gdrive:// URIs) + segmenter
  ``eval_baselines/reelcount_baseline.json``  ? the frozen expected counts (regenerated, never hand-edited)

Usage
-----
Freeze the baseline from the current configured+checkpointed engine (after an intended change, or first run):
    python scripts/nightly_reelcount.py --update-baseline

Nightly check (exit 1 on regression; writes output/nightly_reelcount/<date>.md; emails if --email):
    python scripts/nightly_reelcount.py --email

Exit codes: 0 pass · 1 regression (reel-count changed / a video errored / quality dropped) · 2 operational
(manifest/videos missing, nothing ran) · 3 no baseline · 4 stale (corpus/segmenter changed ? re-freeze).
"""
from __future__ import annotations

import argparse
import datetime as _dt
import json
import os
import sys
from dataclasses import dataclass, field
from typing import Any, Dict, List, Optional, Tuple

REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if REPO_ROOT not in sys.path:
    sys.path.insert(0, REPO_ROOT)

# billing is pure (typing only) ? safe to import at module top.
from backend import billing  # noqa: E402

DEFAULT_MANIFEST = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_manifest.json")
DEFAULT_BASELINE = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_baseline.json")
DEFAULT_REPORT_DIR = os.path.join(REPO_ROOT, "output", "nightly_reelcount")

#: Quality metrics we report + (when labels exist) gate ? higher is better, small tolerance.
QUALITY_HIGHER: Tuple[str, ...] = ("tol_f1", "f1@0.5", "f1@0.25")
DEFAULT_QUALITY_TOL = 0.02     # quality is float/labeled ? a looser tripwire than the exact reel-count
DEFAULT_FX_INR_PER_USD = 83.0
#: Fallback machine price if the manifest doesn't pin one (T4 on n1-standard-8, asia-south1, ~mid-2026).
```

```
DEFAULT_MACHINE_USD_PER_HR = 0.35


@dataclass(frozen=True)
class Tolerances:
    quality_tol: float = DEFAULT_QUALITY_TOL

    def to_dict(self) -> Dict[str, Any]:
        return {"quality_tol": self.quality_tol}


# ------------------------------------------------------------------------ #
# Pure cost math (wraps backend.billing ? no IO).
# ------------------------------------------------------------------------ #
def cost_report(billed_seconds: float, machine_usd_per_hr: float,
                fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
    """Run cost from billed wall-seconds × machine rate ? USD + INR (``backend.billing`` arithmetic).

    ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes ``--vm-uptime-seconds``;
    otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays for boot/install)."""
    rate_per_sec = float(machine_usd_per_hr) / 3600.0
    usage = {billing.WALL_SECONDS: float(billed_seconds)}
    rates = {billing.WALL_SECONDS: rate_per_sec}
    usd, breakdown = billing.compute_cost_usd(usage, rates)
    return {
        "billed_seconds": round(float(billed_seconds), 1),
        "gpu_seconds": round(float(gpu_seconds), 1),
        "machine_usd_per_hr": float(machine_usd_per_hr),
        "usd": usd,
        "inr": billing.to_inr(usd, fx_inr_per_usd),
        "breakdown": breakdown,
    }


# ------------------------------------------------------------------------ #
# Pure summary + comparison core (no IO ? unit-testable with plain dicts).
# ------------------------------------------------------------------------ #
def summarize_results(run_results: List[Dict[str, Any]], segmenter: str,
                      cost: Dict[str, Any]) -> Dict[str, Any]:
    """Compact, comparable snapshot from per-video run results.

    Each run result: ``{name, reel_count, ok, error, quality: {tol_f1,...}|None, wall_seconds}``.
    A video that errored keeps ``reel_count=None`` + ``ok=False`` (so the diff flags it, never hides it)."""
    per_video: Dict[str, Any] = {}
    for r in run_results:
        per_video[r["name"]] = {
            "reel_count": r.get("reel_count"),
            "ok": bool(r.get("ok", False)),
            "error": r.get("error"),
            "quality": r.get("quality"),
        }
    return {"segmenter": segmenter, "cost": cost, "per_video": per_video,
            "n": len(per_video)}


@dataclass
class Finding:
    video: str
    metric: str                 # "reel_count" | "error" | a quality key | "segmenter" | "corpus"
```

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    kind: str                   # "regression" | "improvement" | "stale"
    baseline: Optional[float] = None
    current: Optional[float] = None
    detail: str = ""

    def message(self) -> str:
        if self.kind == "stale":
            return f"[STALE] {self.metric}: {self.detail}"
        if self.metric == "error":
            return f"[REGRESSION] {self.video}: pipeline ERROR ? {self.detail}"
        b = "?" if self.baseline is None else f"{self.baseline:g}"
        c = "?" if self.current is None else f"{self.current:g}"
        tag = "REGRESSION" if self.kind == "regression" else "improvement"
        return f"[{tag}] {self.video}/{self.metric}: {b} ? {c}"


@dataclass
class NightlyResult:
    findings: List[Finding] = field(default_factory=list)
    added: List[str] = field(default_factory=list)
    removed: List[str] = field(default_factory=list)

    @property
    def regressions(self) -> List[Finding]:
        return [f for f in self.findings if f.kind == "regression"]

    @property
    def improvements(self) -> List[Finding]:
        return [f for f in self.findings if f.kind == "improvement"]

    @property
    def stale(self) -> List[Finding]:
        return [f for f in self.findings if f.kind == "stale"]

    def overall_status(self) -> str:
        if self.regressions:
            return "REGRESSION"
        if self.stale:
            return "STALE"
        return "PASS"

    def exit_code(self) -> int:
        if self.regressions:
            return 1
        if self.stale:
            return 4
        return 0


def compare(baseline: Dict[str, Any], current: Dict[str, Any], tols: Tolerances) -> NightlyResult:
    """Diff current run vs frozen baseline.

    * Segmenter changed ? STALE (numbers not comparable; re-freeze).
    * Corpus (video set) changed ? STALE + report added/removed; still diff the INTERSECTION.
    * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR = regression;
      quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
    """
    res = NightlyResult()
    if baseline.get("segmenter") != current.get("segmenter"):
        res.findings.append(Finding("", "segmenter", "stale",
                                    detail=f"{baseline.get('segmenter')} ? {current.get('segmenter')}"))
```

```
        return res

    b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
    res.added = sorted(set(c_pv) - set(b_pv))
    res.removed = sorted(set(b_pv) - set(c_pv))
    if res.added or res.removed:
        res.findings.append(Finding("", "corpus", "stale",
                                    detail=f"added={res.added} removed={res.removed}"))

    for v in sorted(set(b_pv) & set(c_pv)):
        b, c = b_pv[v], c_pv[v]
        # 1) pipeline must still succeed
        if not c.get("ok", False) or c.get("reel_count") is None:
            res.findings.append(Finding(v, "error", "regression",
                                        detail=str(c.get("error") or "no reel_count produced")))
            continue
        # 2) reel count ? EXACT
        bc, cc = b.get("reel_count"), c.get("reel_count")
        if bc is not None and cc is not None and cc != bc:
            res.findings.append(Finding(v, "reel_count", "regression", float(bc), float(cc)))
        # 3) quality metrics ? tolerance (only if both sides have them)
        bq, cq = b.get("quality") or {}, c.get("quality") or {}
        for m in QUALITY_HIGHER:
            bv, cv = bq.get(m), cq.get(m)
            if bv is None or cv is None:
                continue
            if cv < bv - tols.quality_tol:
                res.findings.append(Finding(v, m, "regression", bv, cv))
            elif cv > bv + tols.quality_tol:
                res.findings.append(Finding(v, m, "improvement", bv, cv))
    return res


# --------------------------------------------------------------------------- #
# Report formatting (pure).
# --------------------------------------------------------------------------- #
def _q(qd: Optional[Dict[str, Any]], key: str) -> str:
    if not qd or qd.get(key) is None:
        return "?"
    return f"{qd[key]:.3f}"


def format_report(baseline: Dict[str, Any], current: Dict[str, Any],
                  result: NightlyResult, date: str, head: str) -> str:
    status = result.overall_status()
    badge = {"PASS": "? PASS", "REGRESSION": "? REGRESSION",
             "STALE": "? STALE (re-freeze the baseline)"}[status]
    cost = current.get("cost", {})
    L: List[str] = [
        f"# Nightly engine regression (reel-count + quality + cost) ? {date}",
        "",
        f"**Status: {badge}**  ·  exit code {result.exit_code()}  ·  segmenter `{current.get('segmenter')}`",
        "",
        f"- HEAD `{head}`  ·  baseline `{baseline.get('git_commit', '?')}` (frozen {baseline.get('generated_at', '?')})",
        f"- **Run cost: ${cost.get('usd', 0):.4f} (?{cost.get('inr', 0):.2f})**  ·  "
        f"billed {cost.get('billed_seconds', 0):.0f}s @ ${cost.get('machine_usd_per_hr', 0)}/hr",
        "",
    ]
    if result.regressions:
        L += ["## ? Regressions", ""] + [f"- {f.message()}" for f in result.regressions]
```

```
+ [""]
    if result.stale:
        L += ["## ? Stale / not comparable", ""] + [f"- {f.message()}" for f in result.stale]
        L += ["- Action: `python scripts/nightly_reelcount.py --update-baseline` (after confirming the change is intended).", ""]

    # Per-video table.
    b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
    L += ["## Per-video", "",
          "| video | reels (base?now) | tol_f1 (base?now) | f1@0.5 | status |",
          "|---|---|---|---|---|"]
    for v in sorted(set(b_pv) | set(c_pv)):
        b, c = b_pv.get(v, {}), c_pv.get(v, {})
        bc = b.get("reel_count")
        cc = c.get("reel_count")
        rc = f"{'?' if bc is None else bc}?{'ERROR' if not c.get('ok', True) else ('?' if cc is None else cc)}"
        bq, cq = b.get("quality"), c.get("quality")
        tf = f"{_q(bq, 'tol_f1')}?{_q(cq, 'tol_f1')}"
        f5 = f"{_q(bq, 'f1@0.5')}?{_q(cq, 'f1@0.5')}"
        vstatus = "?" if any(f.video == v and f.kind == "regression" for f in result.findings) else "?"
        L.append(f"| {v} | {rc} | {tf} | {f5} | {vstatus} |")
    L.append("")

    if result.improvements:
        L += ["_Improvements over baseline (re-freeze to ratchet up):_"] \
            + [f"- {f.message()}" for f in result.improvements] + [""]
    L += ["---",
          "_Full configured+checkpointed pipeline on the real model (GCP GPU VM). Tripwire only ? does "
          "not change the promotion gate. Complements PR #202's cached-trajectory CPU re-score._"]
    return "\n".join(L)


# --------------------------------------------------------------------------- #
# IO shell ? running the configured pipeline per video (needs backend + GPU + video).
# --------------------------------------------------------------------------- #
def _load_gts(labels_ref: Optional[str]) -> Optional[List[Tuple[float, float]]]:
    """Load ground-truth rally intervals from a golden-features JSON (``gts`` key) or a rallies CSV
    (``start,end`` columns / pairs). Returns None when no labels are configured."""
    if not labels_ref:
        return None
    from backend.storage import fetch
    local = fetch(labels_ref)
    if local.endswith(".json"):
        with open(local, encoding="utf-8") as fh:
            data = json.load(fh)
        return [(float(s), float(e)) for s, e in (data.get("gts") or [])]
    # CSV: header start,end (or first two numeric columns)
    import csv
    out: List[Tuple[float, float]] = []
    with open(local, encoding="utf-8") as fh:
        for row in csv.reader(fh):
            try:
                out.append((float(row[0]), float(row[1])))
            except (ValueError, IndexError):
                continue
    return out


def _segments_to_intervals(segs: List[Dict[str, Any]]) -> List[Tuple[float, float]]:
```

```
    """``(start, end)`` pairs from segmenter result dicts, tolerant of the two key conventions in the
    codebase (``start_time``/``end_time`` ? motion/DB shape ? or ``start``/``end``). Used only for the
    optional quality metrics; the reel COUNT is ``len(segs)`` regardless, so a key mismatch degrades
    quality scoring gracefully (skipped) without ever miscounting reels."""
    out: List[Tuple[float, float]] = []
    for s in segs:
        start = s.get("start_time", s.get("start"))
        end = s.get("end_time", s.get("end"))
        if start is not None and end is not None:
            out.append((float(start), float(end)))
    return out


def run_one_video(video: Dict[str, Any], segmenter: str, config: Any,
                  rerun_on_mismatch: bool = True,
                  baseline_count: Optional[int] = None) -> Dict[str, Any]:
    """Run the configured pipeline on one video ? reel count (+ quality vs labels). Lazy backend imports.

    The re-run-once guard (the one known non-determinism ? a transient ``add_play_segment``?None drop,
    database.py): if the count != the frozen baseline, re-process the video ONCE; a transient drop won't
    reproduce, a real change will."""
    import random
    import time

    from backend.eval import rally_seg_eval
    from backend.infrastructure.database import Database
    from backend.pipeline.segmenters import segmenter_registry
    from backend.results import Failure
    from backend.storage import fetch
    from backend.utils.hashing import compute_video_id

    name = video["name"]
    try:
        local = fetch(video["uri"])
    except Exception as e:  # noqa: BLE001 ? surface, never hide
        return {"name": name, "reel_count": None, "ok": False, "error": f"fetch failed: {e}",
                "quality": None, "wall_seconds": 0.0}

    gts = _load_gts(video.get("labels_uri"))
    seg_cls = segmenter_registry.get(segmenter)
    if not seg_cls:
        return {"name": name, "reel_count": None, "ok": False,
                "error": f"segmenter '{segmenter}' not registered", "quality": None, "wall_seconds": 0.0}

    def _process(db_path: str) -> Tuple[Optional[int], Optional[List[Tuple[float, float]]], float, Optional[str]]:
        random.seed(0)  # close motion.py's unseeded metadata RNG so secondary fields are reproducible too
        db = Database(db_path)
        video_id = compute_video_id(local)
        t0 = time.monotonic()
        result = seg_cls(db, config).process_video(video_id, local)
        wall = time.monotonic() - t0
        if isinstance(result, Failure):
            return None, None, wall, getattr(result, "message", "segmenter failed")
        segs = result.value if hasattr(result, "value") else result
        intervals = _segments_to_intervals(segs)
```

```
            return len(segs), intervals, wall, None

        import tempfile
        total_wall = 0.0
        with tempfile.TemporaryDirectory(prefix="nightly_reel_") as td:
            count, intervals, wall, err = _process(os.path.join(td, "run1.db"))
            total_wall += wall
            if err is not None:
                return {"name": name, "reel_count": None, "ok": False, "error": err,
                        "quality": None, "wall_seconds": round(total_wall, 2)}
            # re-run-once guard for the transient DB-drop flake
            if rerun_on_mismatch and baseline_count is not None and count != baseline_count:
                count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
                total_wall += wall2
                if err2 is None and count2 is not None:
                    count, intervals = count2, intervals2  # the reproduced (or corrected) value

            quality = None
            if gts and intervals is not None:
                row = rally_seg_eval.score_intervals(intervals, gts)
                quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25", "tol_precision", "tol_recall")
                           if k in row}

        return {"name": name, "reel_count": count, "ok": True, "error": None,
                "quality": quality, "wall_seconds": round(total_wall, 2)}


def load_manifest(path: str) -> Dict[str, Any]:
    with open(path, encoding="utf-8") as fh:
        return json.load(fh)


def run_corpus(manifest: Dict[str, Any], baseline: Optional[Dict[str, Any]],
               rerun_on_mismatch: bool = True) -> Tuple[List[Dict[str, Any]], float]:
    """Run every manifest video through the configured pipeline. Returns (run_results, total_wall_s)."""
    from backend.config import load_config

    config = load_config()
    # apply manifest config overrides onto the typed config (dotted keys)
    for dotted, val in (manifest.get("config_overrides") or {}).items():
        _set_dotted(config, dotted, val)
    segmenter = manifest.get("segmenter") or getattr(getattr(config, "indexing", None), "default_segmenter", "motion")
    b_pv = (baseline or {}).get("per_video", {})
    results: List[Dict[str, Any]] = []
    total_wall = 0.0
    for v in manifest.get("videos", []):
        base_count = (b_pv.get(v["name"]) or {}).get("reel_count")
        r = run_one_video(v, segmenter, config, rerun_on_mismatch, base_count)
        total_wall += float(r.get("wall_seconds") or 0.0)
        results.append(r)
    return results, total_wall


def _set_dotted(config: Any, dotted: str, value: Any) -> None:
    """Best-effort dotted-path set onto a pydantic config (e.g. 'indexing.motion_threshold')."""
    parts = dotted.split(".")
    obj = config
    for p in parts[:-1]:
        obj = getattr(obj, p, None)
        if obj is None:
```

```
            return
    try:
        setattr(obj, parts[-1], value)
    except Exception:  # noqa: BLE001 ? overrides are best-effort; a bad key shouldn't crash the run
        pass


def _git_head() -> str:
    import subprocess
    try:
        out = subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
                             capture_output=True, text=True, timeout=10)
        return out.stdout.strip() or "?"
    except Exception:  # noqa: BLE001
        return "?"


def load_baseline(path: str) -> Optional[Dict[str, Any]]:
    if not os.path.exists(path):
        return None
    with open(path, encoding="utf-8") as fh:
        return json.load(fh)


def save_baseline(path: str, snapshot: Dict[str, Any]) -> None:
    out = dict(snapshot)
    out["generated_at"] = _dt.date.today().isoformat()
    out["git_commit"] = _git_head()
    out["_doc"] = ("Frozen nightly reel-count baseline. Regenerate with `python scripts/nightly_reelcount.py "
                   "--update-baseline` after an INTENDED engine change or when adding a video. Tied to the "
                   "configured+checkpointed engine + the manifest corpus.")
    out.pop("cost", None)  # cost is per-run, not part of the frozen baseline
    os.makedirs(os.path.dirname(path), exist_ok=True)
    tmp = path + ".tmp"
    with open(tmp, "w", encoding="utf-8") as fh:
        json.dump(out, fh, indent=2, sort_keys=True)
        fh.write("\n")
    os.replace(tmp, path)


def build_parser() -> argparse.ArgumentParser:
    p = argparse.ArgumentParser(description="Nightly E2E engine regression: reel-count + quality + cost.")
    p.add_argument("--manifest", default=DEFAULT_MANIFEST)
    p.add_argument("--baseline", default=DEFAULT_BASELINE)
    p.add_argument("--report-dir", default=DEFAULT_REPORT_DIR)
    p.add_argument("--quality-tol", type=float, default=DEFAULT_QUALITY_TOL)
    p.add_argument("--vm-uptime-seconds", type=float, default=None,
                   help="authoritative VM uptime (boot?delete) for the cost line; "
                        "defaults to the in-process pipeline wall time (a lower bound)")
    p.add_argument("--no-rerun-guard", action="store_true",
                   help="disable the re-run-once-on-mismatch guard (the DB-drop flake mitigation)")
    p.add_argument("--update-baseline", action="store_true",
                   help="freeze the current reel counts + quality as the new baseline instead of checking")
    p.add_argument("--email", action="store_true", help="email the report (see scripts/nightly_email.py)")
    p.add_argument("--email-to", default=None, help="override the report recipient")
    p.add_argument("--date", default=None, help="override the report date (YYYY-MM-DD; for tests)")
    p.add_argument("--no-report", action="store_true", help="skip writing the dated
```

```
report file")
    return p


def main(argv: Optional[List[str]] = None) -> int:
    import logging
    logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
    log = logging.getLogger("nightly_reelcount")
    args = build_parser().parse_args(argv)
    tols = Tolerances(quality_tol=args.quality_tol)
    date = args.date or _dt.date.today().isoformat()

    if not os.path.exists(args.manifest):
        log.error("manifest not found: %s", args.manifest)
        return 2
    manifest = load_manifest(args.manifest)
    baseline = load_baseline(args.baseline)

    try:
        run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not args.no_rerun_guard)
    except Exception as e:  # noqa: BLE001 ? operational failure (deps/config) ? exit 2, never a false "pass"
        log.error("run failed: %s", e)
        return 2
    if not run_results:
        log.error("no videos in manifest %s", args.manifest)
        return 2

    billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else total_wall
    cost = cost_report(
        billed_seconds=billed,
        machine_usd_per_hr=float(manifest.get("machine_usd_per_hr", DEFAULT_MACHINE_USD_PER_HR)),
        fx_inr_per_usd=float(manifest.get("fx_inr_per_usd", DEFAULT_FX_INR_PER_USD)),
        gpu_seconds=total_wall,
    )
    segmenter = manifest.get("segmenter", "motion")
    snapshot = summarize_results(run_results, segmenter, cost)
    snapshot["git_commit"] = _git_head()
    snapshot["generated_at"] = date

    if args.update_baseline:
        save_baseline(args.baseline, snapshot)
        log.info("baseline frozen ? %s", args.baseline)
        for r in run_results:
            log.info("  %s: reels=%s ok=%s", r["name"], r.get("reel_count"), r.get("ok"))
        return 0

    if baseline is None:
        log.error("no baseline at %s ? freeze one with --update-baseline (on the configured engine).", args.baseline)
        return 3

    result = compare(baseline, snapshot, tols)
    report = format_report(baseline, snapshot, result, date, snapshot["git_commit"])

    if not args.no_report:
        os.makedirs(args.report_dir, exist_ok=True)
        md_path = os.path.join(args.report_dir, f"{date}.md")
        with open(md_path, "w", encoding="utf-8") as fh:
            fh.write(report + "\n")
        with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8")
```

```
as fh:
            json.dump({"date": date, "status": result.overall_status(), "exit_code": result.exit_code(),
                       "snapshot": snapshot, "regressions": [f.message() for f in result.regressions]},
                      fh, indent=2, sort_keys=True)
        log.info("report ? %s", md_path)

    print(report)

    if args.email:
        try:
            from scripts.nightly_email import send_report
            subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count engine regression {date}"
            sent = send_report(subject, report, to=args.email_to)
            log.info("email %s", "sent" if sent else "skipped (no creds ? report still written)")
        except Exception as e:  # noqa: BLE001 ? a broken mailer must NOT mask the regression verdict
            log.error("email failed (report still written): %s", e)

    code = result.exit_code()
    if code == 1:
        log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions), args.report_dir)
    elif code == 4:
        log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-baseline")
    else:
        log.info("nightly reel-count check PASSED  (cost $%.4f / ?%.2f)", cost["usd"], cost["inr"])
    return code


if __name__ == "__main__":
    raise SystemExit(main())
```

4. user (2026-06-22T09:42:16.431Z)

```
"""Email the nightly regression report ? stdlib SMTP, creds from env or GCP Secret Manager.

Used by ``scripts/nightly_reelcount.py --email`` (and reusable by any nightly). Deliberately
dependency-light (stdlib ``smtplib`` + ``email.message``) and **graceful**: if no SMTP password can
be resolved it logs + returns ``False`` (the report is still written/printed ? a broken mailer must
never mask a regression).

Credential resolution (first hit wins), so it works on a GCP VM, locally, or in tests:
  ? Recipient:  ``--email-to`` arg  ?  ``NIGHTLY_EMAIL_TO``  ?  ``avi.dullu@gmail.com``
  ? SMTP host:  ``NIGHTLY_SMTP_HOST``  (default ``smtp.gmail.com``)   · port ``NIGHTLY_SMTP_PORT`` (587, STARTTLS)
  ? SMTP user:  ``NIGHTLY_SMTP_USER``  (default = recipient)
  ? SMTP pass:  ``NIGHTLY_SMTP_PASS``  ?  Secret Manager ``NIGHTLY_SMTP_SECRET``
                (``projects/<p>/secrets/<name>/versions/latest``; via the google-cloud-secret-manager
                 lib if installed, else the ``gcloud`` CLI). A Gmail **app password** is the simplest.

Set the secret once (owner):  echo -n '<app-password>' | gcloud secrets create nightly-smtp-pass --data-file=-
```

```
and on the VM:  export NIGHTLY_SMTP_SECRET=projects/<proj>/secrets/nightly-smtp-pass/versions/latest
                export NIGHTLY_SMTP_USER=<your-gmail> NIGHTLY_EMAIL_TO=avi.dullu@gmail.com
"""
from __future__ import annotations

import logging
import os
from email.message import EmailMessage
from typing import Optional

log = logging.getLogger("nightly_email")

DEFAULT_TO = "avi.dullu@gmail.com"
DEFAULT_HOST = "smtp.gmail.com"
DEFAULT_PORT = 587


def build_message(subject: str, body_md: str, from_addr: str, to_addr: str) -> EmailMessage:
    """A text/plain email carrying the markdown report (renders fine in any client). Pure ? testable."""
    msg = EmailMessage()
    msg["Subject"] = subject
    msg["From"] = from_addr
    msg["To"] = to_addr
    msg.set_content(body_md)
    return msg


def resolve_secret(env: Optional[dict] = None) -> Optional[str]:
    """Resolve the SMTP password from env or GCP Secret Manager. Returns None if unresolvable."""
    env = env if env is not None else os.environ
    direct = env.get("NIGHTLY_SMTP_PASS")
    if direct:
        return direct
    secret_ref = env.get("NIGHTLY_SMTP_SECRET")
    if not secret_ref:
        return None
    # Prefer the client lib; fall back to the gcloud CLI (no extra dependency required).
    try:
        from google.cloud import secretmanager  # type: ignore
        client = secretmanager.SecretManagerServiceClient()
        return client.access_secret_version(name=secret_ref).payload.data.decode("utf-8")
    except Exception:  # noqa: BLE001 ? lib absent or auth issue ? try the CLI
        pass
    try:
        import subprocess
        name = secret_ref.split("/secrets/", 1)[1].split("/", 1)[0] if "/secrets/" in secret_ref else secret_ref
        out = subprocess.run(["gcloud", "secrets", "versions", "access", "latest", "--secret", name],
                             capture_output=True, text=True, timeout=30)
        return out.stdout.strip() if out.returncode == 0 and out.stdout.strip() else None
    except Exception as e:  # noqa: BLE001
        log.warning("secret-manager resolution failed: %s", e)
        return None


def send_report(subject: str, body_md: str, to: Optional[str] = None,
                env: Optional[dict] = None) -> bool:
```

```
    """Send the report. Returns True if sent, False if skipped (no creds) ? never raises on missing creds."""
    env = env if env is not None else os.environ
    to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
    host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
    port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
    user = env.get("NIGHTLY_SMTP_USER") or to_addr
    password = resolve_secret(env)
    if not password:
        log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ? email skipped.")
        return False

    import smtplib
    msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
    with smtplib.SMTP(host, port, timeout=60) as s:
        s.starttls()
        s.login(user, password)
        s.send_message(msg)
    log.info("emailed report to %s", to_addr)
    return True

```

5. user (2026-06-22T09:42:17.081Z)

```
#!/usr/bin/env bash
# Startup-script for the NIGHTLY engine-regression VM. Runs UNATTENDED on boot, then DELETES the VM
# (cost rule: a running GPU VM is the only real cost). Survives SSH drops (it's a startup-script, not
# an interactive session). Reads its config from instance metadata (set by launch.sh).
#
# Flow:  Ops Agent  ?  pull repo @ HEAD from GCS  ?  build venv + WASB env  ?  stage weights  ?
#        run scripts/nightly_reelcount.py on the gs:// golden clips (emails the report) ?
#        upload report to GCS  ?  SELF-DELETE (even on failure, via the EXIT trap).
#
# Metadata keys (launch.sh sets these): repo-tgz-uri, git-sha, gcs-bucket, report-prefix,
#   weights-uri, email (true/false). NOT verified in CI ? dry-run via launch.sh on a real VM first.
set -uo pipefail
exec > >(tee -a /var/log/nightly-regression.log) 2>&1
echo "===== nightly-regression startup $(date -u +%FT%TZ) ====="
BOOT_EPOCH=$(date +%s)

md() { curl -s -H "Metadata-Flavor: Google" "http://metadata.google.internal/computeMetadata/v1/$1"; }
NAME=$(md instance/name)
ZONE=$(md instance/zone | awk -F/ '{print $NF}')
REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
BUCKET=$(md instance/attributes/gcs-bucket || true)
REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
DO_EMAIL=$(md instance/attributes/email || echo true)
SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
SMTP_USER=$(md instance/attributes/smtp-user || true)
EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
DATE=$(date -u +%F)

# ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM behind.
cleanup() {
```

```
  echo "===== self-delete $NAME ($ZONE) ====="
  gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
    echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete $NAME --zone=$ZONE -q"
}
trap cleanup EXIT

fail_email() {  # best-effort failure alert (the report email won't have fired on an early crash)
  echo "RUN FAILED ? attempting a failure alert"
  if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
    cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
    NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER" NIGHTLY_EMAIL_TO="$EMAIL_TO" \
    python - <<PY || true
from scripts.nightly_email import send_report
send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
            "The nightly engine-regression run failed on the VM before producing a report.\n"
            "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
PY
  fi
}
trap 'fail_email' ERR

# 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
  sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent install failed (non-fatal)"

# 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the worker SA).
mkdir -p ~/rally && cd ~/rally
gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
tar -xzf /tmp/rally.tgz -C ~/rally
echo "$GIT_SHA" > ~/rally/GIT_SHA

# 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per deploy/gcp/README §4).
sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
export TMPDIR=/scratch 2>/dev/null || true
./deploy/digitalocean/bootstrap.sh --gpu
. .venv/bin/activate
pip install -e '.[storage-gcs]'
bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned nonzero ? continuing"

# 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
if [ -n "$WEIGHTS_URI" ]; then
  mkdir -p ~/models/WASB-SBDT/pretrained_weights
  gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
  export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
fi

# 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM uptime so far.
UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
set +e
NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER" NIGHTLY_EMAIL_TO="$EMAIL_TO" \
  python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
RUN_RC=$?
```

```
set -e
echo "runner exit code: $RUN_RC  (0 pass / 1 regression / 4 stale)"

# 6) Upload the dated report to GCS for the record (alongside the email).
if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
  gcloud storage cp -r "output/nightly_reelcount/$DATE."* "gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
    gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/" || true
fi

echo "===== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
trap - ERR   # disable the failure-email trap; the EXIT (self-delete) trap stays
# the EXIT trap deletes the VM

```

6. user (2026-06-22T09:42:18.013Z)

```
#!/usr/bin/env bash
# Launch ONE nightly engine-regression run on an ephemeral GCP GPU VM, then return. The VM runs
# run_on_vm.sh as its startup-script (regression ? email ? SELF-DELETE), so this just stages the repo
# and kicks the VM off. Driver-agnostic: call it from Cloud Scheduler?Cloud Run job, a cron on any
# always-on box, the owner's machine, or a Claude routine. Needs gcloud auth + the repo checkout.
#
#   bash deploy/gcp/nightly_regression/launch.sh                 # L4, stages HEAD, kicks the VM
#   RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash deploy/gcp/nightly_regression/launch.sh
#
# ? Run this MANUALLY once as a dry-run (watch the serial log) before wiring a scheduler ? the VM-side
# flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-output <name> --zone=<z>.
set -uo pipefail

PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
BUCKET="${RALLY_BUCKET:-khelsutra-rally-corpus}"
SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
GPU="${RALLY_GPU:-l4}"
NAME="${RALLY_VM_NAME:-rally-nightly-$(date -u +%Y%m%d)}"
REF="${RALLY_REF:-HEAD}"
WEIGHTS_URI="${RALLY_WEIGHTS_URI:-gs://$BUCKET/weights/wasb_badminton_best.pth.tar}"
REPORT_PREFIX="${RALLY_REPORT_PREFIX:-nightly/reports}"
DO_EMAIL="${RALLY_EMAIL:-true}"
SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-pass/versions/latest}"
SMTP_USER="${NIGHTLY_SMTP_USER:-}"
EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
STARTUP="$HERE/run_on_vm.sh"
export CLOUDSDK_CORE_DISABLE_PROMPTS=1

ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a asia-southeast1-b asia-southeast1-c}"
if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else MACHINE="n1-standard-8"; ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi

# 1) Stage the repo @ REF to GCS (no GitHub auth on the VM; the worker SA can read the bucket).
SHA="$(git rev-parse "$REF")"
TGZ_URI="gs://$BUCKET/nightly/repo/${SHA}.tgz"
```

```
echo "== staging repo $SHA -> $TGZ_URI =="
TMP_TGZ="$(mktemp --suffix=.tgz)"
git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
rm -f "$TMP_TGZ"

# 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config metadata.
ERR="$(mktemp)"
for Z in $ZONES; do
  echo "== creating $NAME ($GPU) in $Z =="
  if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
       --machine-type="$MACHINE" "${ACCEL[@]}" \
       --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
       --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-platform-release \
       --boot-disk-size=120GB --boot-disk-type=pd-balanced \
       --service-account="$SA" --scopes=cloud-platform \
       --metadata-from-file=startup-script="$STARTUP" \
       --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
       2>"$ERR"; then
    echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and self-delete."
    echo "   serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z --project=$PROJECT"
    echo "   if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z -q"
    rm -f "$ERR"; exit 0
  fi
  if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo "  (no $GPU capacity in $Z ? next zone)"; continue; fi
  echo "  create failed:"; tail -4 "$ERR"; rm -f "$ERR"; exit 1
done
echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
rm -f "$ERR"; exit 1

```

7. user (2026-06-22T09:42:18.683Z)

```
#!/usr/bin/env bash
# Wire the nightly engine-regression as a CLOUD-NATIVE, scale-to-zero job:
#   Cloud Scheduler (cron)  ?  Cloud Run JOB (the launcher image)  ?  launch.sh
#     ?  ephemeral GPU VM (regression + email + SELF-DELETE).
# The only always-on cost is $0 (Cloud Run jobs + Scheduler are scale-to-zero; the GPU VM is ephemeral).
#
# ? OWNER-RUN, ONCE, after a successful manual `bash deploy/gcp/nightly_regression/launch.sh` dry-run.
# NOT verified in CI (no GCP/GPU here). Prereqs: GPU quota (README §2), the golden clips + weights in
# gs://<bucket>, and the SMTP secret (see README "Enable").
#
#   bash deploy/gcp/nightly_regression/register_scheduler.sh           # build + deploy + schedule (03:00 IST)
#   RALLY_CRON='0 3 * * *' RALLY_TZ='Asia/Kolkata' bash .../register_scheduler.sh
set -euo pipefail

PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
REGION="${RALLY_RUN_REGION:-asia-south1}"
JOB="${RALLY_RUN_JOB:-nightly-regression-launcher}"
SCHED="${RALLY_SCHED_NAME:-nightly-regression}"
CRON="${RALLY_CRON:-0 3 * * *}"          # 03:00 daily
TZ="${RALLY_TZ:-Asia/Kolkata}"
```

```
SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
IMAGE="${RALLY_LAUNCHER_IMAGE:-${REGION}-docker.pkg.dev/${PROJECT}/rally/nightly-launcher:latest}"
REPO_ROOT="$(git rev-parse --show-toplevel)"
export CLOUDSDK_CORE_DISABLE_PROMPTS=1

echo "== enable APIs (run, scheduler, build, artifact registry) =="
gcloud services enable run.googleapis.com cloudscheduler.googleapis.com \
  cloudbuild.googleapis.com artifactregistry.googleapis.com --project="$PROJECT"

echo "== ensure the Artifact Registry repo 'rally' exists =="
gcloud artifacts repositories describe rally --location="$REGION" --project="$PROJECT" >/dev/null 2>&1 || \
  gcloud artifacts repositories create rally --repository-format=docker --location="$REGION" --project="$PROJECT"

echo "== build + push the launcher image from the repo root =="
( cd "$REPO_ROOT" && gcloud builds submit --project="$PROJECT" --tag "$IMAGE" \
    --gcs-source-staging-dir="gs://${RALLY_BUILD_BUCKET:-khelsutra-rally-corpus}/cloudbuild" \
    -f deploy/gcp/nightly_regression/Dockerfile.launcher . )

echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU VM) =="
gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
  --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
  --set-env-vars="RALLY_GPU=${RALLY_GPU:-l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"

echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
RUN_URI="https://${REGION}-run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
gcloud scheduler jobs describe "$SCHED" --location="$REGION" --project="$PROJECT" >/dev/null 2>&1 && \
  gcloud scheduler jobs delete "$SCHED" --location="$REGION" --project="$PROJECT" --quiet || true
gcloud scheduler jobs create http "$SCHED" --project="$PROJECT" --location="$REGION" \
  --schedule="$CRON" --time-zone="$TZ" \
  --uri="$RUN_URI" --http-method=POST \
  --oauth-service-account-email="$SA"

cat <<EOF

? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
   IAM the launcher SA needs (grant once if missing):
     roles/compute.instanceAdmin.v1  (create/delete the GPU VM)
     roles/iam.serviceAccountUser    (attach $SA to the VM)
     roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
   Trigger now to test:  gcloud run jobs execute $JOB --region=$REGION --project=$PROJECT
   Remove:               gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud run jobs delete $JOB --region=$REGION
EOF

```

8. assistant (2026-06-22T09:42:20.986Z)

Now let me read the remaining files.

9. user (2026-06-22T09:42:22.291Z)

```
# Tiny launcher image for the nightly engine-regression Cloud Run JOB. It is NOT the GPU workload ?
# it only stages the repo + creates the ephemeral GPU VM (which self-deletes). Scale-to-zero: the job
```

```
# runs for a few seconds nightly, then exits. Base = google/cloud-sdk (gcloud + git preinstalled).
FROM gcr.io/google.com/cloudsdktool/cloud-sdk:slim
WORKDIR /app
# The repo is the build context (run `gcloud builds submit` from the repo root, see register_scheduler.sh).
COPY . /app
ENTRYPOINT ["bash", "deploy/gcp/nightly_regression/launch.sh"]

```

10. user (2026-06-22T09:42:23.295Z)


```
{
  "_doc": "Nightly E2E engine reel-count manifest (scripts/nightly_reelcount.py). The configured+checkpointed pipeline runs on each video on a GCP GPU VM; the reel count + quality metrics are asserted vs eval_baselines/reelcount_baseline.json. ADD A VIDEO = append one row + re-freeze: `python scripts/nightly_reelcount.py --update-baseline` (commit the updated baseline). URIs may be gs:// (recommended, co-located with the VM), gdrive://, or local paths. labels_uri (optional) is a golden-features JSON (gts key) or a start,end CSV ? present => quality metrics (tol_f1/f1@0.5/f1@0.25) are scored too. EDIT the example rows below to your actual short (~5-6 min) OWNED, minors-free clips before enabling.",
  "segmenter": "wasb_hybrid",
  "detector_impl": "native",
  "config_overrides": {
    "indexing.default_segmenter": "wasb_hybrid",
    "indexing.detector_impl": "native"
  },
  "machine_usd_per_hr": 0.70,
  "fx_inr_per_usd": 83.0,
  "videos": [
    {
      "name": "GX030094",
      "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX030094.mp4",
      "labels_uri": "gs://khelsutra-rally-corpus/nightly/labels/GX030094_golden_features.json"
    },
    {
      "name": "Badminton_BXH_2",
      "uri": "gs://khelsutra-rally-corpus/nightly/videos/Badminton_BXH_2.mp4",
      "labels_uri": "gs://khelsutra-rally-corpus/nightly/labels/Badminton_BXH_2_golden_features.json"
    },
    {
      "name": "GX010128",
      "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX010128.mp4",
      "labels_uri": "gs://khelsutra-rally-corpus/nightly/labels/GX010128_golden_features.json"
    }
  ]
}

```


11. assistant (2026-06-22T09:42:26.044Z)

Now I'll analyze the bash scripts in detail, focusing on the critical safety and correctness issues.

12. user (2026-06-22T09:42:26.970Z)

```
./deploy/gcp/nightly_regression/launch.sh
./deploy/gcp/nightly_regression/register_scheduler.sh
./deploy/gcp/nightly_regression/run_on_vm.sh
```

13. assistant (2026-06-22T09:42:29.162Z)

Now let me perform a detailed analysis of the bash safety issues:

14. user (2026-06-22T09:42:30.585Z)

```
8:#        upload report to GCS  ?  SELF-DELETE (even on failure, via the EXIT trap).
12:set -uo pipefail
37:trap cleanup EXIT
46:send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
52:trap 'fail_email' ERR
67:./deploy/digitalocean/bootstrap.sh --gpu
86:set -e
96:trap - ERR   # disable the failure-email trap; the EXIT (self-delete) trap stays
97:# the EXIT trap deletes the VM
```

15. assistant (2026-06-22T09:42:32.526Z)

Let me analyze the trap interactions more carefully:

16. user (2026-06-22T09:42:33.555Z)

```
12      set -uo pipefail
13      exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14      echo "===== nightly-regression startup $(date -u +%FT%TZ) ====="
15      BOOT_EPOCH=$(date +%s)
16
17      md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18      NAME=$(md instance/name)
19      ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20      REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21      GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22      BUCKET=$(md instance/attributes/gcs-bucket || true)
23      REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24      WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25      DO_EMAIL=$(md instance/attributes/email || echo true)
26      SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27      SMTP_USER=$(md instance/attributes/smtp-user || true)
28      EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29      DATE=$(date -u +%F)
30
31      # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32      cleanup() {
33        echo "===== self-delete $NAME ($ZONE) ====="
34        gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35          echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36      }
37      trap cleanup EXIT
38
39      fail_email() {  # best-effort failure alert (the report email won't have fired on an
early crash)
40        echo "RUN FAILED ? attempting a failure alert"
41        if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42          cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43          NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44          python - <<PY || true
45      from scripts.nightly_email import send_report
46      send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47                  "The nightly engine-regression run failed on the VM before producing a
report.\n"
48                  "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
```

```
PY
  fi
}
trap 'fail_email' ERR

# 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
  sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent install failed (non-fatal)"

# 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the worker SA).
mkdir -p ~/rally && cd ~/rally
gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
tar -xzf /tmp/rally.tgz -C ~/rally
echo "$GIT_SHA" > ~/rally/GIT_SHA

# 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per deploy/gcp/README §4).
sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
export TMPDIR=/scratch 2>/dev/null || true
./deploy/digitalocean/bootstrap.sh --gpu
. .venv/bin/activate
pip install -e '.[storage-gcs]'
bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned nonzero ? continuing"

# 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
if [ -n "$WEIGHTS_URI" ]; then
  mkdir -p ~/models/WASB-SBDT/pretrained_weights
  gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
  export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
fi

# 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM uptime so far.
UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
set +e
NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER" NIGHTLY_EMAIL_TO="$EMAIL_TO" \
  python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
RUN_RC=$?
set -e
echo "runner exit code: $RUN_RC  (0 pass / 1 regression / 4 stale)"

# 6) Upload the dated report to GCS for the record (alongside the email).
if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
  gcloud storage cp -r "output/nightly_reelcount/$DATE."* "gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
    gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/" || true
fi

echo "===== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
trap - ERR   # disable the failure-email trap; the EXIT (self-delete) trap stays
# the EXIT trap deletes the VM
```

17. assistant (2026-06-22T09:42:35.542Z)

Now let me check launch.sh for metadata escaping and zone sweep correctness:

18. user (2026-06-22T09:42:37.290Z)

```
echo "== staging repo $SHA -> $TGZ_URI =="
TMP_TGZ="$(mktemp --suffix=.tgz)"
git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
rm -f "$TMP_TGZ"

# 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config metadata.
ERR="$(mktemp)"
for Z in $ZONES; do
  echo "== creating $NAME ($GPU) in $Z =="
  if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
       --machine-type="$MACHINE" "${ACCEL[@]}" \
       --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
       --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-platform-release \
       --boot-disk-size=120GB --boot-disk-type=pd-balanced \
       --service-account="$SA" --scopes=cloud-platform \
       --metadata-from-file=startup-script="$STARTUP" \
       --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
       2>"$ERR"; then
    echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and self-delete."
    echo "   serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z --project=$PROJECT"
    echo "   if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z -q"
    rm -f "$ERR"; exit 0
  fi
  if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo "  (no $GPU capacity in $Z ? next zone)"; continue; fi
  echo "  create failed:"; tail -4 "$ERR"; rm -f "$ERR"; exit 1
done
echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
rm -f "$ERR"; exit 1
```

19. assistant (2026-06-22T09:42:39.795Z)

Now let me check the Scheduler OAuth and Run job setup:

20. user (2026-06-22T09:42:40.383Z)

```
echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU VM) =="
gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
  --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
  --set-env-vars="RALLY_GPU=${RALLY_GPU:-l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"

echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
RUN_URI="https://${REGION}-run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
gcloud scheduler jobs describe "$SCHED" --location="$REGION" --project="$PROJECT" >/dev/null 2>&1 && \
  gcloud scheduler jobs delete "$SCHED" --location="$REGION" --project="$PROJECT" --quiet || true
gcloud scheduler jobs create http "$SCHED" --project="$PROJECT" --location="$REGION" \
  --schedule="$CRON" --time-zone="$TZ" \
  --uri="$RUN_URI" --http-method=POST \
  --oauth-service-account-email="$SA"
```

```
    cat <<EOF

    ? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
       IAM the launcher SA needs (grant once if missing):
         roles/compute.instanceAdmin.v1  (create/delete the GPU VM)
         roles/iam.serviceAccountUser    (attach $SA to the VM)
         roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
       Trigger now to test:  gcloud run jobs execute $JOB --region=$REGION --project=$PROJECT
       Remove:               gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud run jobs delete $JOB --region=$REGION
    EOF
```

21. assistant (2026-06-22T09:42:42.414Z)

Let me check for any issues with env var expansion in metadata:

22. user (2026-06-22T09:42:43.522Z)

```
       --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
       2>"$ERR"; then
    echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and self-delete."
    echo "   serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z --project=$PROJECT"
    echo "   if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z -q"
    rm -f "$ERR"; exit 0
```

23. assistant (2026-06-22T09:42:45.645Z)

Now let me verify the trap logic carefully:

24. user (2026-06-22T09:42:48.046Z)

```
Setting up traps...
About to fail...
ERR trap fired
After false
EXIT trap fired
```

25. assistant (2026-06-22T09:42:49.948Z)

Let me test the ERR trap behavior with `set -e`:

26. user (2026-06-22T09:42:53.787Z)

```
Exit code 1
=== Test 1: ERR trap interaction with set +e / set -e ===
Step 1: running with set -uo pipefail
Step 2: set +e, now running false
ERR trap fired
Step 3: false returned, set -e next
Step 4: set -e active, about to call a function
ERR trap fired
EXIT trap fired
```

27. assistant (2026-06-22T09:42:56.006Z)

Now let me verify the metadata escaping potential:

28. user (2026-06-22T09:43:00.358Z)

```
Testing metadata expansion:
--metadata=key="value,with,commas and spaces"

If we used this in: gcloud compute instances create --metadata=key="value,with,commas and
spaces"
The shell would parse it as one argument because we quoted it.

BUT if SMTP_SECRET or other env var contains a comma:
With our quoted approach: --metadata=smtp-secret="projects/p/secrets/name,sneaky=val"
This would be passed as a single value to gcloud.
```

29. assistant (2026-06-22T09:43:02.859Z)

```
Let me check what happens if an early step fails in run_on_vm.sh:
```

30. user (2026-06-22T09:43:06.147Z)

```
Setting up traps...
Step 1: mkdir and cd
Step 2: This fails early...
fail_email (ERR): send failure email
Step 3: never reached because set -uo pipefail
cleanup (EXIT): self-delete
```

31. assistant (2026-06-22T09:43:09.006Z)

```
Good. Now let me test a critical issue: what if BOOT_EPOCH fails to set?
```

32. user (2026-06-22T09:43:12.515Z)

```
BOOT_EPOCH=1782121392
Later calculation...
UPTIME=0
Setting BOOT_EPOCH to empty...
```

33. assistant (2026-06-22T09:43:14.749Z)

```
Good?empty BOOT_EPOCH causes arithmetic to fail silently (treats empty as 0). Now let me test
metadata value escaping more carefully:
```

34. user (2026-06-22T09:43:24.496Z)

```
Case 1: Quoted value with comma
--metadata=foo="gs://bucket/prefix,sneaky=val"

Full metadata string:
--metadata=repo-tgz-uri="gs://bucket/tgz",git-sha="abc123",gcs-bucket="bucket",report-
prefix="nightly/reports",weights-uri="gs://bucket/w.pth",email="true",smtp-
secret="projects/p/secrets/nightly-smtp-pass/versions/latest",smtp-user="",email-
to="avi.dullu@gmail.com"

If REPORT_PREFIX='reports,prod', the string becomes:
--metadata=report-prefix="reports,prod",weights-uri=...
gcloud would parse: report-prefix="reports, prod",weights-uri
Notice the comma INSIDE the quotes is OK, but if the QUOTE is missing:

If we passed UNQUOTED: --metadata=smtp-secret=val,sneaky=evil
gcloud would parse TWO pairs: smtp-secret=val AND sneaky=evil
```

35. assistant (2026-06-22T09:43:26.569Z)

```
Now let me test a critical issue: what if the metadata key is missing?
```

36. user (2026-06-22T09:43:31.582Z)

REPO_TGZ='Simulating: curl -s -H 'Metadata-Flavor: Google' 'http://metadata.google.internal/computeMetadata/v1/instance/attributes/repo-tgz-uri'' (empty if key missing)

37. assistant (2026-06-22T09:43:35.580Z)

Let me look at the actual robustness of metadata reads:

38. user (2026-06-22T09:43:37.245Z)

```
md() { curl -s -H "Metadata-Flavor: Google" "http://metadata.google.internal/computeMetadata/v1/$1"; }
NAME=$(md instance/name)
ZONE=$(md instance/zone | awk -F/ '{print $NF}')
REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
BUCKET=$(md instance/attributes/gcs-bucket || true)
REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
DO_EMAIL=$(md instance/attributes/email || echo true)
SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
SMTP_USER=$(md instance/attributes/smtp-user || true)
EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
DATE=$(date -u +%F)

# ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM behind.
```

39. assistant (2026-06-22T09:43:39.949Z)

Perfect. Now let me analyze the Python code for issues with the cost calculation:

40. user (2026-06-22T09:43:40.851Z)

<tool_use_error>InputValidationError: Read failed due to the following issue:
The parameter `offset` type is expected as `number` but provided as `array`</tool_use_error>

41. user (2026-06-22T09:43:43.501Z)

```
def cost_report(billed_seconds: float, machine_usd_per_hr: float,
                fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
    """Run cost from billed wall-seconds × machine rate ? USD + INR (``backend.billing`` arithmetic).

    ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes ``--vm-uptime-seconds``;
    otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays for boot/install)."""
    rate_per_sec = float(machine_usd_per_hr) / 3600.0
    usage = {billing.WALL_SECONDS: float(billed_seconds)}
    rates = {billing.WALL_SECONDS: rate_per_sec}
    usd, breakdown = billing.compute_cost_usd(usage, rates)
    return {
        "billed_seconds": round(float(billed_seconds), 1),
        "gpu_seconds": round(float(gpu_seconds), 1),
        "machine_usd_per_hr": float(machine_usd_per_hr),
        "usd": usd,
        "inr": billing.to_inr(usd, fx_inr_per_usd),
        "breakdown": breakdown,
    }


```

42. assistant (2026-06-22T09:43:45.507Z)

Now let me check for issues with the UPTIME calculation and edge cases:

43. user (2026-06-22T09:43:51.447Z)

```
=== Scenario 1: What if repo pull fails early ===
BOOT_EPOCH=1782121431
UPTIME=0

=== Scenario 2: What if BOOT_EPOCH gets overwritten? ===
Initial BOOT_EPOCH=1782121431

=== Scenario 3: What if date +%s fails on line 80? ===
UPTIME=0 (relies on date succeeding both times)
```

44. assistant (2026-06-22T09:43:53.579Z)

Now let me verify the Cloud Run Job and Scheduler setup for correctness:

45. user (2026-06-22T09:44:01.008Z)

```
=== Cloud Scheduler Setup Analysis ===

Line 45 RUN_URI:
  https://${REGION}-
run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run

This URI should trigger the Cloud Run Job 'execute' endpoint.

Line 51 --oauth-service-account-email=$SA
  This tells the Scheduler to use OIDC auth with the service account.

POTENTIAL ISSUE:
  The Cloud Run Job needs to INVOKE, but the URI is the admin 'execute' endpoint.
  The SA must have either:
    1. roles/run.admin (full control)
    2. roles/run.invoker (execute only)

  Line 59 says 'roles/run.invoker + roles/cloudscheduler.jobRunner' needed.
  This is CORRECT but incomplete documentation:
    - run.invoker: invoke the Run Job
    - cloudscheduler.jobRunner: let Scheduler invoke jobs

However, the documentation doesn't mention that the SA running the
Scheduler trigger ALSO needs to have 'roles/run.jobs.run' to invoke the job
if using the :run endpoint. This may be OK if the Scheduler SA == the Job SA.
```

46. assistant (2026-06-22T09:44:02.905Z)

Let me verify one more critical issue: does the Cloud Run Job properly set env vars?

47. user (2026-06-22T09:44:04.560Z)

```
<tool_use_error>InputValidationError: Read failed due to the following issue:
The parameter `offset` type is expected as `number` but provided as `array`</tool_use_error>
```

48. user (2026-06-22T09:44:07.989Z)

```
39      echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU
VM) =="
40      gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
41        --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
42        --set-env-vars="RALLY_GPU=${RALLY_GPU:-
```

l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
43

49. assistant (2026-06-22T09:44:10.256Z)

Now let me check the Dockerfile.launcher to see if it properly passes env vars:

50. user (2026-06-22T09:44:10.942Z)

```
# Tiny launcher image for the nightly engine-regression Cloud Run JOB. It is NOT the GPU workload ?
# it only stages the repo + creates the ephemeral GPU VM (which self-deletes). Scale-to-zero: the job
# runs for a few seconds nightly, then exits. Base = google/cloud-sdk (gcloud + git preinstalled).
FROM gcr.io/google.com/cloudsdktool/cloud-sdk:slim
WORKDIR /app
# The repo is the build context (run `gcloud builds submit` from the repo root, see register_scheduler.sh).
COPY . /app
ENTRYPOINT ["bash", "deploy/gcp/nightly_regression/launch.sh"]
```

51. assistant (2026-06-22T09:44:13.440Z)

Perfect. Now let me compile all my findings. Let me do one more check on the trap/ERR interaction:

52. user (2026-06-22T09:44:19.318Z)

Simulating mkdir -p ~/rally
Now simulating a failure at line 60 (gcloud storage cp):
FAIL_EMAIL: Attempted to send failure email
After failure (should not reach here due to set -uo pipefail)
CLEANUP: Would delete VM

53. assistant (2026-06-22T09:44:21.192Z)

Perfect! Now I have all the information I need. Let me compile my findings:

54. user (2026-06-22T09:45:25.703Z)

Structured output provided successfully

55. assistant (2026-06-22T09:45:35.876Z)

Summary

I've completed a thorough security and correctness review of the nightly reel-count regression engine. The analysis identified **3 CRITICAL defects** that could leave a billing GPU VM alive or cause silent failures, **4 MEDIUM issues** affecting reliability and cost accuracy, and **3 LOW-severity findings**.

Critical Findings (Must Fix Before Production)

1. **Unguarded metadata key reads** (lines 18-28): NAME and ZONE could be empty, causing self-delete to fail silently.
2. **Missing REPO_TGZ validation** (line 20, 60-61): If the metadata key is absent, the pipeline silently fails with a confusing error and no alert to the owner.
3. **UPTIME calculation underflow** (lines 15, 80): If `date +%s` fails, BOOT_EPOCH defaults to unset, causing UPTIME to incorrectly calculate the full wall time instead of uptime.

Medium Issues

- **Metadata value escaping** (launch.sh:53): Special characters in SMTP_SECRET or other paths could break gcloud parsing.
- **Premature ERR trap disabling** (run_on_vm.sh:96): The trap is disabled before the GCS upload, so upload failures don't trigger failure alerts.
- **Zone sweep uniqueness** (launch.sh:43-64): Multiple VMs with the same name could be created if the loop somehow retries.
- **Scheduler IAM incomplete** (register_scheduler.sh:56-59): Missing documented permission for the Cloud Run :run endpoint invocation.

All findings have been returned with specific line numbers, severity levels, and remediation guidance. The Python code (nightly_reelcount.py, nightly_email.py) is well-structured and safe?the risks are exclusively in the bash orchestration and GCP configuration.